

TECHNICAL DOCUMENTATION

MQTT Virtual Device Dashboard

MQTTVirtualDeviceDashboard.jsx

Complete implementation reference for all features simulating a STLA M6 connected vehicle TBOX

COMPONENT	ROUTE	FRAMEWORK
MQTTVirtualDeviceDashboard.jsx	/mqtt-virtual-device	React 18 + Framer Motion

DOC DATE	SPEC VERSION	SOURCE REV
15 April 2026	SB Interface V5.0	Rev 1.1 — 15-Apr-2026

SOURCE REFERENCE DOCUMENTS

MQTT_Virtual_Device_3012 — Rev 1.1 — 15-Apr-2026

TE 01 _ STLA M6 Connectivity South Bound Interface Specification V5.0

Jeep_Proto/*.proto — stla.cvip.jeep.*

certificates: ca.crt, client.crt, client.key

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The `MQTTVirtualDeviceDashboard.jsx` is a fully self-contained React 18 component that **simulates a connected vehicle TBOX (Telematics Box / Dongle) device** communicating over the MQTT protocol with protobuf-encoded messages over the CVIP (Connected Vehicle India Platform). It is embedded inside the Rule Engine Dashboard at `/mqtt-virtual-device`.

⚡ KEY FEATURE

No real MQTT broker is required. All payloads are mock-simulated as structured JSON matching the actual protobuf schema from the source Node.js scripts. The component can be used entirely offline for diagnostic and testing purposes.

► Dashboard Capabilities

- ✓ Configure full device identity: VIN, IMEI, TBOX Serial, Broker URL, Protocol Version
- ✓ Simulate MQTT connect / disconnect lifecycle with visual connection state machine
- ✓ Publish 6 categories of MQTT messages: Vehicle Telemetry, Location, Events, Alerts, Trips, Device Join
- ✓ Simulate 8 types of incoming server commands and send back command responses with correlationId
- ✓ View 6 live telemetry gauges with animated spring-count transitions and trend arrows
- ✓ Browse a scrollable, filterable message log with JSON + Hex payload expansion
- ✓ Auto Drive mode — continuous GPS drift + telemetry stream every 3 seconds
- ✓ Export full log as JSON for offline analysis

► Technology Stack

LIBRARY	VERSION	USAGE
<code>react</code>	18.x	Component framework; <code>useReducer</code> state management
<code>framer-motion</code>	^11.18.2	All physics-based spring animations (replaces "Antigravity" engine)
<code>uuid</code>	^13.0.0	UUID v4 generation for messageId, correlationId, logId
Node.js <code>mqtt</code>	5.8.0	Source reference (not bundled); all MQTT logic is simulated
Node.js <code>protobufjs</code>	^8.0.0	Source reference (not bundled); mock hex payloads generated

2 Source Reference Files Read

2.1 MQTT Virtual Device Node.js Scripts — Rev 1.1 (15-Apr-2026)

DIRECTORY	SCRIPT FILE	IMPLEMENTED FEATURE	IN DASHBOARD
deviceJoin/	deviceJoin.js	Device handshake / registration / topic setup	✓
vehicleTelemetry/	vehicleTelemetryFuelLevel.js	CAN Fuel Level — identifier 854, PDU 0x64...	✓
vehicleTelemetry/	engineSpeed.js	CAN Engine Speed (RPM)	✓
vehicleTelemetry/	engineWaterTemp.js	CAN Engine Water Temperature (°C)	✓
vehicleTelemetry/	vehicleTelemetryBatteryVoltage.js	CAN Battery Voltage (V)	✓
vehicleTelemetry/	odoMeter.js	CAN Odometer (km)	✓
locationTelemetry/	locationTelemetry.js	GPS lat, long, alt, course, signal quality	✓
events/	start_event.js	Ignition START event	✓
events/	stop_event.js	Ignition STOP event	✓
events/	remoteBlinkerOnOffCommand.js	Remote blinker ON / OFF events	✓
events/	remoteDoorLockUnlockCommand.js	Door lock / unlock events	✓
events/	remoteHonkEventStatus.js	Remote honk event status	✓
events/	tboxInFullSleepStatus.js	TBOX full sleep entry event	✓
events/	vehicleMobilizationStatus.js	Vehicle mobilization status	✓

DIRECTORY	SCRIPT FILE	IMPLEMENTED FEATURE	IN DASHBOARD
alerts/	speedAlertWithValue.js	Speed alert with vehicleSpeed value + GPS	✓
alerts/	harshAccelerationAlert.js	Harsh acceleration alert	✓
alerts/	harshBreakingALert.js	Harsh braking alert	✓
alerts/	sharpTurnsAlert.js	Sharp turns alert	✓
alerts/	engineIdlingAlert.js	Engine idling alert	✓
alerts/	parkingDisturbanceAlert.js	Parking disturbance alert	✓
alerts/	towAwayAlert.js	Tow-away alert	✓
alerts/	dongleStatusAlert.js	Dongle health / status alert	✓
alerts/	remoteImmobilizationAlert.js	Remote immobilization alert	✓
alerts/	remoteMobilizationAlert.js	Remote mobilization alert	✓
trips/	tripStart.js	Trip start lifecycle + start GPS	✓
trips/	tripCurrent.js	Trip progress / current data	✓
trips/	tripEnd.js	Trip end + total distance / duration	✓
command/	remotedoorlockcommand.js	Door lock command response	✓
command/	remoteDoorUnLockCommandResponse.js	Door unlock command response	✓
command/	remoteBlinkerOnCommand.js	Blinker ON command response	✓
command/	remoteBlinkerOffCommand.js	Blinker OFF command response	✓
command/	remoteHonkCommand.js	Honk command response	✓

2.2 STLA M6 SB Interface Specification V5.0 Sections Used

SECTION	TOPIC	API INTERFACE ID	IMPLEMENTED
§6.2	MQTT Protocol — QoS 1, MQTT v3.1.1, TLS 1.2	—	✓
§6.2.1	Connection flags: Clean Session=false, ClientId=VIN/IMEI	—	✓
§6.2.4	Keep Alive: 30s (configurable)	—	✓ Documented
§6.2.5	LWT Topic: <code>/dongle/{VIN}/MQTTPROTOBUF/deviceJoined</code>	—	✓ Documented
§7.3.4.1	deviceJoined / deviceJoinedRsp message format	SB-DONGLE_ACTIVATION-007	✓
§7.3.4.1.3	Regex validation rules for deviceJoined parameters	—	✓
§8.1	Topic Nomenclature: <code>/dongle/{VIN}/MQTTPROTOBUF/{type}</code>	—	✓ All 13 topics
§9	Dongle Commands (command / commandResponse topics)	SB-DONGLE_COMMANDS-001	✓
§9.2	Fetch Device Logs command	SB-DONGLE_COMMANDS-003	✓
§9.4	Dongle State Update command	SB-DONGLE_COMMANDS-004	✓ Simulated
§10.7	Remote Blinker ON Command	SB-REMOTE_COMMANDS-001	✓
§10.8	Remote Blinker OFF Command	SB-REMOTE_COMMANDS-002	✓

SECTION	TOPIC	API INTERFACE ID	IMPLEMENTED
§10.9	Remote Door Lock Command	SB-REMOTE_COMMANDS-003	✓
§10.10	Remote Door Unlock Command	SB-REMOTE_COMMANDS-004	✓
§10.11	Remote Honk Command	SB-REMOTE_COMMANDS-005	✓
§10.12	Remote command failure codes (26 failure codes)	SB-REMOTE_COMMANDS-006	✓ Documented
§11	FOTA — fotaCommand / fotaCommandResponse topics	SB-FOTA_COMMANDS-001	✓ Simulated
§12.2	Vehicle Telemetry stream (CAN data)	—	✓
§12.3	Location Telemetry stream (GPS)	—	✓
§13.1	Vehicle Alerts (10 types)	—	✓
§13.2	Vehicle Events (10 types)	—	✓
§13.11	Trip Start payload	SB-TRIP_TELEMETRY-006	✓
§13.12	Trip Current/Progress payload	SB-TRIP_TELEMETRY-006	✓
§13.13	Trip End payload	SB-TRIP_TELEMETRY-006	✓

3 Section 1 — Device Identity Configuration

Source: `.env` from MQTT Virtual Device Rev 1.1 + `deviceJoin.js` lines 19–28

3.1 Configurable Fields

FIELD	DEFAULT VALUE	.ENV VARIABLE	SPEC REFERENCE
VIN	T123ZTZT867657777	VEHICLE_ID	§7.3.4.1 vehicleId, regex: <code>^[a-zA-Z0-9]{17,20}\$</code>
IMEI	356741396798769	IMEI	§7.2.1 15-digit IMEI, TAC="35674136"
TBOX Serial	T123ZTZT1396798769	TBOX_SERIAL	§7.3.4.1 tboxSerialNum
Client ID	T123ZTZT867657777	CLIENT_ID	§6.2.1 — VIN after provisioning
Message ID	UUID v4	MESSAGE_ID	§8 — "UUIDs based on current timestamp"
Correlation ID	UUID v4	CORRELATION_ID	§4 Glossary — "message ID of corresponding command"
Broker URL	cvipiot-preprod.fca-india.com:18883	BROKER_URL	§6.2.6 TLS 1.2 Preprod
Protocol Version	2.0.0	—	§8 "version: 2.0.0" — all payloads

3.2 Broker Options (Dropdown)

LABEL	FULL URL	PORT
FCA India Preprod	mqtt://cvipiot-preprod.fca-india.com	18883
Citroen Preprod	mqtt://lb2.cvip-preprod.citroen.in	48883

3.3 Action Buttons

- ✓  **Generate UUIDs** — Refreshes both `messageId` and `correlationId` with new UUID v4 values
- ✓  **Save Config** — Persists draft form values to global React state + fires toast notification

4 Section 2 — MQTT Connection Panel

Source: `deviceJoin.js` `mqtt.connect()` options + Spec §6.2

4.1 Connection State Machine

STATE	COLOR	DOT ANIMATION	TRIGGER
DISCONNECTED	Gray #52525b	Static	Initial state / Disconnect button
CONNECTING	Amber #f59e0b	Scale pulse loop, 0.9s	Click ► CONNECT
CONNECTED	Green #10b981	Slow pulse loop, 2.0s	~800ms after CONNECTING
ERROR	Red #f43f5e	Rapid shake oscillation	Simulated network failure

4.2 Live Connection Stats

STAT	DISPLAY	UPDATE TRIGGER
Published	Integer counter	Every successful onPublish()
Received	Integer counter	Every simulated SUB message
Uptime	HH:MM:SS format	setInterval 1000ms while CONNECTED
Last Pub	HH:MM:SS.ms	Most recent publish timestamp

4.3 MQTT Parameters (from source + Spec)

```
clientId:      CLIENT_ID (VIN/IMEI – per §6.2.1)
protocolVersion: 4 (MQTTv3.1.1 – per §6.2)
clean:         false (persist session – per §6.2.1 "Clean Session = 0")
keepAlive:     30 seconds (§6.2.4: 10 minutes in spec, 30s simulated)
QoS:           1 (at least once – per §6.2.2)
LWT topic:     /dongle/{VIN}/MQTTPROTOBUF/deviceJoined (§6.2.5)
TLS version:   1.2 (§6.2.6 Preprod environment)
```

5 Section 3 — Publish Panel (All 6 Tabs)

5.1 Vehicle Telemetry Tab

Source: `vehicleTelemetry/*.js` — Spec §12.2 — Proto: `stla.cvip.jeep.CANDataMessage`

Published Topic: `/dongle/{VIN}/MQTTPROTOBUF/vehicleTelemetry`

► CAN Signal Types (Dropdown)

SIGNAL	CAN IDENTIFIER	UNIT	PDU DEFAULT	SOURCE FILE
Fuel Level	854	%	0x6400000000000000	vehicleTelemetryFuelLevel.js
Engine Speed	100	RPM	Configurable decimal input	engineSpeed.js
Engine Water Temp	100	°C	Configurable decimal input	engineWaterTemp.js
Battery Voltage	100	V	Configurable decimal input	vehicleTelemetryBatteryVoltage.js
Odometer	100	km	Configurable integer input	odoMeter.js

► Full Payload Structure (from vehicleTelemetryFuelLevel.js)

```
{
  "messageId": "dd52bab1-c64a-431b-a945-1602f26e9f4e",
  "eTboxApplicationState": "customer",
  "tboxEsimState": "normal_sim",
  "tboxOperatingState": "normal",
  "version": "2.0.0",
  "timeStamp": { "seconds": 1750000000, "nanos": 0 },
  "canDataPayload": {
    "canData": {
      "canDataPacket": [{
        "identifier": 854,
        "dlc": 8,
        "cData": [{
          "pdu": "0x6400000000000000",
          "timestamp": { "seconds": 1750000000, "nanos": 0 }
        }]
      }]
    }
  }
}
```

```
}]  
}  
}  
}
```

► Additional Configurable Fields

FIELD	OPTIONS	PROTO ENUM
App State	customer / provisioned / authorized_customer / provisioned_authorized	<code>eTboxApplicationState</code>
eSIM State	normal_sim / no_sim / roaming_sim	<code>tboxEsimState</code>
Operating State	normal / sleep / full_sleep / boot	<code>tboxOperatingState</code>
Interval (sec)	1–3600 (number)	Auto-repeat timer
Auto Toggle	☞ Auto / ■ Stop	setInterval / clearInterval

5.2 Location Telemetry Tab

Source: `locationTelemetry/locationTelemetry.js` — Spec §12.3 — Proto: `stla.cvip.jeeep.LocationTelemetryMessage`

Published Topic: `/dongle/{VIN}/MQTTPROTOBUF/locationTelemetry`

FIELD	TYPE	DEFAULT (CHENNAI)	SPEC FIELD
GPS Latitude	double	13.084534	<code>gpsLat</code>
GPS Longitude	double	80.270718	<code>gpsLong</code>
GPS Altitude	float	20.0 m	<code>gpsAlt</code>
GPS Course Angle	float	180.0°	<code>gpsCourseAngle</code>
GPS Signal Quality	float	97.0	<code>gpsSignalQuality</code>
GPS Fixed Status	int32	1 (Fixed)	<code>gpsFixedStatus</code>
Speed	float	25 km/h	<code>speed</code>
Signal Strength	int32	30	<code>signalStrength</code>

A 400×180px dark grid map renders an animated amber pin that tracks lat/long inputs in real-time using Framer Motion `useSpring` (stiffness: 120, damping: 18). The pin shows current coordinates as text, and moves smoothly when values change.

5.3 Events Tab

Source: `events/*.js` — Spec §13.2 — Proto: `stla.cvip.jeep.EventMessage`

Published Topic: `/dongle/{VIN}/MQTTPROTOBUF/events`

EVENT TYPE	SOURCE SCRIPT	SPEC REFERENCE
IgnitionStatus: START	<code>start_event.js</code>	§13.2 IgnitionPropertyType.START
IgnitionStatus: STOP	<code>stop_event.js</code>	§13.2 IgnitionPropertyType.STOP
RemoteBlinkerOn	<code>remoteBlinkerOnOffCommand.js</code>	§10.7 RemoteBlinkerOnCommand
RemoteBlinkerOff	<code>remoteBlinkerOnOffCommand.js</code>	§10.8 RemoteBlinkerOffCommand
RemoteDoorLock	<code>remoteDoorLockUnlockCommand.js</code>	§10.9 RemoteDoorLockCommand
RemoteDoorUnlock	<code>remoteDoorLockUnlockCommand.js</code>	§10.10 RemoteDoorUnLockCommand
RemoteHonkStatus	<code>remoteHonkEventStatus.js</code>	§10.11 RemoteHonkCommand
VehicleMobilizationStatus	<code>vehicleMobilizationStatus.js</code>	§13.2
TBoxInFullSleep	<code>tboxInFullSleepStatus.js</code>	§11.1.4.1 Full Sleep
VehicleMobilization	<code>vehicleMobilizationStatus.js</code>	§13.2

5.4 Alerts Tab

Source: `alerts/*.js` — Spec §13.1 — Proto: `stla.cvip.jeep.AlertsMessage`

Published Topic: `/dongle/{VIN}/MQTTPROTOBUF/alerts`

ALERT TYPE	SOURCE SCRIPT	EXTRA FIELDS
SpeedAlert	<code>speedAlertWithValue.js</code>	vehicleSpeed (km/h) + GPS lat/long
HarshAcceleration	<code>harshAccelerationAlert.js</code>	alertState, GPS
HarshBraking	<code>harshBreakingALert.js</code>	alertState, GPS
SharpTurns	<code>sharpTurnsAlert.js</code>	alertState

ALERT TYPE	SOURCE SCRIPT	EXTRA FIELDS
EngineIdling	<code>engineIdlingAlert.js</code>	alertState
ParkingDisturbance	<code>parkingDisturbanceAlert.js</code>	alertState
TowAway	<code>towAwayAlert.js</code>	alertState, isLive
DongleStatus	<code>dongleStatusAlert.js</code>	alertState
RemoteImmobilization	<code>remoteImmobilizationAlert.js</code>	alertState
RemoteMobilization	<code>remoteMobilizationAlert.js</code>	alertState

► **Alert State Options (from proto enum)**

`alsAlert` (active alarm) | `alsNormal` (cleared) | `alsUnknown` (indeterminate)

5.5 Trips Tab

Source: `trips/*.js` — Spec §13.11–13.13 — Proto: `stla.cvip.jeep.TripMessage`

SUB-TAB	PUBLISHED TOPIC	KEY FIELDS
Trip Start	<code>/dongle/{VIN}/MQTTPROTOBUF/tripStart</code>	tripId (date-based int64), startTime, startDate, gpsLat, gpsLong, gpsAlt, gpsCourseAngle, gpsFixedStatus
Trip Current	<code>/dongle/{VIN}/MQTTPROTOBUF/tripCurrent</code>	tripId, distanceSoFar (km), currentSpeed (km/h), currentTimeStamp
Trip End	<code>/dongle/{VIN}/MQTTPROTOBUF/tripEnd</code>	tripId, endGpsLat, endGpsLong, totalDistance (km), duration (minutes)

TRIP ID GENERATION

Auto-generated as `DDMMYY + "03"` encoded as int64. E.g., for 15-Apr-2026 → `150426`. Source: `tripStart.js` line ~20.

5.6 Device Join Tab

Source: `deviceJoin/deviceJoin.js` — Spec §7.3.4.1 — SB-DONGLE_ACTIVATION-007

Published Topic: `/dongle/{VIN}/MQTTPROTOBUF/deviceJoined`

Auto-Subscribe on Publish: `/dongle/{VIN}/MQTTPROTOBUF/deviceJoinedRsp` and `/dongle/{VIN}/MQTTPROTOBUF/command`

```

{
  "vehicleId":      "T123ZTZT867657777",
  "tboxSerialNum":  "T123ZTZT1396798769",
  "imeiNo":         "356741396798769",
  "protocolVersion": "2.0.0",
  "tboxOperatingState": "NORMAL",
  "tboxApplicationState": "PROVISIONED",
  "tboxeSimState":    "NORMAL_SIM",
  "NAD_SW_Version":   "ND0.00.11",
  "MCU_SW_Version":   "MD0.00.03"
}

```

► **Regex Validation Rules — Spec §7.3.4.1.3**

FIELD	REGEX / ALLOWED VALUES
vehicleId	<code>^[a-zA-Z0-9]{17,20}\$</code>
tboxSerialNum	<code>^[a-zA-Z0-9]{13}\$</code>
protocolVersion	<code>([0-9][.])([0-9][.])[0-9]</code>
imeiNo	<code>^[0-9]{15}\$</code>
tboxOperatingState	<code>normal disconnected NORMAL DISCONNECTED</code>
tboxApplicationState	<code>factory provisioned authorized customer</code> (any case)
tboxeSimState	<code>normal_sim NORMAL_SIM</code>
NAD_SW_Version	<code>^[a-zA-Z0-9]{3}\.[a-zA-Z0-9]{2}\.[a-zA-Z0-9]{2}\$</code>
MCU_SW_Version	<code>^[a-zA-Z0-9]{3}\.[a-zA-Z0-9]{2}\.[a-zA-Z0-9]{2}\$</code>

Source: `command/*.js` — Spec §9 (Dongle Commands) — SB-REMOTE_COMMANDS-001 to 006

Subscribe Topic: `/dongle/{VIN}/MQTTPROTOBUF/command`

Publish Response: `/dongle/{VIN}/MQTTPROTOBUF/commandResponse`

COMMAND	SOURCE FILE	SB API ID	SUBTYPE
Remote Door Lock	<code>remotedoorlockcommand.js</code>	SB- REMOTE_COMMANDS- 003	RemoteDoorLockCommand
Remote Door Unlock	<code>remoteDoorUnLockCommandResponse.js</code>	SB- REMOTE_COMMANDS- 004	RemoteDoorUnLockCommand
Remote Blinker ON	<code>remoteblinkeroncommand.js</code>	SB- REMOTE_COMMANDS- 001	RemoteBlinkerOnCommand
Remote Blinker OFF	<code>remoteblinkeroffcommand.js</code>	SB- REMOTE_COMMANDS- 002	RemoteBlinkerOffCommand
Remote Honk	<code>remotehonkcommand.js</code>	SB- REMOTE_COMMANDS- 005	RemoteHonkCommand
FOTA Command	fotaCommands/	SB- FOTA_COMMANDS- 001	fotaCommand
OTA Command	—	FOTA extension	otaCommand
Fetch Device Log	fetchDeviceLog/	SB- DONGLE_COMMANDS- 003	FetchLogsCommand

► Command Response Structure — from `remotedoorlockcommand.js`

```
{
  "messageId": "dd52bab1- ...",
  "correlationId": "<extracted-from-command-messageId>",
  "subtype": "RemoteDoorLockCommandResponse",
  "tboxOperatingState": "normal",
  "eTboxApplicationState": "customer",
  "tboxEsimState": "normal_sim",
  "returnCode": "succeeded",           // succeeded | failed
  "commandStatus": "success",         // success | failure
  "version": "2.0.0",
  "timeStamp": { "seconds": ..., "nanos": ... }
}
```

SPEC §10.6 — KEY DESIGN CONSIDERATION

"If the command is received by Dongle **after 60 seconds** of command-initiated timestamp, such commands need to be ignored by Dongle and need not to be processed." The command received-at timestamp is visible in the dashboard for tester awareness.

► Remote Command Failure Codes — Spec §10.12 (26 codes)

CODE	NAME
0	invalidRemoteFailureCode
1	KeyIn_ACC
2	KeyIn_IGN
3	Engine_Running_or_Ready_ToDrive
4	LVBAT_LowVoltage_or_High_Voltage
8	Bounce
9	FotalInProgress
16	authentication_failed
17	authentication_timeout
18	request_failed
23	DoorOpenLock
25	RequestRefused

CODE	NAME
26	RequestFailed

7

Section 5 — Live Telemetry Cards

CARD	SIGNAL SOURCE	UNIT	COLOR	UPDATE SOURCE
Fuel Level	CAN identifier 854	%	■ Amber	Vehicle Telemetry tab publish
Engine Speed	CAN RPM	RPM	■ Emerald	Vehicle Telemetry tab / Auto Drive
Water Temp	CAN temp	°C	■ Rose	Vehicle Telemetry tab / Auto Drive
Battery	CAN voltage	V	■ Sky	Vehicle Telemetry tab
Odometer	CAN distance	km	■ Violet	Vehicle Telemetry tab / Auto Drive
GPS Lat	Location payload	°	■ Teal	Location Telemetry tab / Auto Drive

Per-card features: animated number count; trend arrow ↑/↓/→; dynamic box-shadow glow when value updates.

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Section 6 — Message Log Panel

FEATURE	DETAIL
Direction badges	↑ PUB and ↓ SUB
Status badges	SUCCESS FAILED PENDING
Max entries	200 (older entries evicted from state)
Payload expansion	Click entry → JSON pretty-print + Hex payload (proto mock, first 96 chars)
Filter: Direction	ALL / PUB / SUB toggle buttons
Filter: Message type	Free-text search input
Filter: Status	ALL / SUCCESS / FAILED / PENDING toggle buttons
Clear Log	Animated scatter — entries fly off screen before clearing

FEATURE	DETAIL
Export JSON	Downloads all entries as <code>mqtt_log.json</code> via Blob URL

9 Automation — Simulate Auto Drive

Trigger: "★ Simulate Auto Drive" toggle button (left column). Must be CONNECTED first.

PUBLISHED EVERY 3S	BEHAVIOR
Location Telemetry	gpsLat/Long drifts by $\pm 0.001^\circ$ random walk per tick
Vehicle Telemetry — Speed	Fluctuates 0–140 km/h with sinusoidal drift
Vehicle Telemetry — Fuel	Slowly decreases 0.01% per tick
Vehicle Telemetry — RPM	Proportional to speed with noise
Vehicle Telemetry — Water Temp	Warms up to $\sim 90^\circ\text{C}$ and stabilizes

Pattern: `/dongle/{VIN}/MQTTPROTOBUF/{type}` — Spec §8.1

Publish Topics — Dongle → CVIP

#	TOPIC SUFFIX	SB API ID	TAB/SOURCE	IMPLEMENTED
1	<code>deviceJoined</code>	SB-DONGLE_ACTIVATION-007	Device Join tab	✓
2	<code>vehicleTelemetry</code>	§12.2	Vehicle Tel tab	✓
3	<code>locationTelemetry</code>	§12.3	Location tab	✓
4	<code>events</code>	§13.2	Events tab	✓
5	<code>alerts</code>	§13.1	Alerts tab	✓
6	<code>tripStart</code>	SB-TRIP_TELEMETRY-006	Trips → Start	✓
7	<code>tripCurrent</code>	SB-TRIP_TELEMETRY-006	Trips → Current	✓
8	<code>tripEnd</code>	SB-TRIP_TELEMETRY-006	Trips → End	✓
9	<code>commandResponse</code>	SB-DONGLE_COMMANDS-001	Command Inbox	✓
10	<code>fotaCommandResponse</code>	SB-FOTA_COMMANDS-001	Command Inbox (FOTA)	✓

Subscribe Topics — CVIP → Dongle (Simulated)

#	TOPIC SUFFIX	SB API ID	SIMULATED VIA	IMPLEMENTED
1	<code>deviceJoinedRsp</code>	SB-DONGLE_ACTIVATION-007	Auto-fires 1.2s after deviceJoined publish	✓
2	<code>command</code>	SB-DONGLE_COMMANDS-001	"↓ Simulate" buttons in Command Inbox	✓
3	<code>fotaCommand</code>	SB-FOTA_COMMANDS-001	"↓ FOTA Command" button	✓

11 State Management Architecture

Global state is managed via **React** `useReducer` with a single normalized state tree consisting of 8 top-level slices:

```
const initialState = {
  config:      { vin, imei, tboxSerial, clientId, messageId, correlationId, broker, protocolVersion },
  connection:  { status, publishCount, receiveCount, uptimeSeconds, lastPublishAt, connectedAt },
  telemetry:   { fuelLevel, engineSpeed, engineWaterTemp, batteryVoltage, odometer, gpsLat, gpsLong ... },
  telemetryHistory: { fuelLevel: [...10], engineSpeed: [...10], ... },    // trend arrow computation
  log:         [ { id, timestamp, direction, messageType, topic, status, payloadBytes, ... } ],
  commandInbox: [ { id, commandType, correlationId, receivedAt, responseStatus } ],
  toasts:      [ { id, message, type } ],
  activePublishTab: 'vehicleTelemetry',
  autoDriveActive: false
};
```

► Reducer Actions

ACTION TYPE	EFFECT
<code>SET_CONFIG</code>	Merges new config values from Device Identity panel
<code>SET_CONNECTION</code>	Partial merge of connection state slice
<code>INCREMENT_PUBLISH</code>	publishCount++, updates lastPublishAt
<code>INCREMENT_RECEIVE</code>	receiveCount++, updates lastReceiveAt
<code>SET_TELEMETRY</code>	Updates single telemetry key + appends to history array
<code>SET_TELEMETRY_BULK</code>	Updates multiple telemetry keys at once
<code>ADD_LOG</code>	Prepends entry to log[], max 200 enforced
<code>CLEAR_LOG</code>	Resets log to empty array
<code>ADD_COMMAND</code>	Appends command to commandInbox[]
<code>UPDATE_COMMAND</code>	Updates responseStatus for a specific command
<code>ADD_TOAST</code>	Appends toast to toasts[], max 5
<code>REMOVE_TOAST</code>	Removes toast by id (auto after 3500ms)
<code>SET_ACTIVE_TAB</code>	Updates activePublishTab
<code>SET_AUTO_DRIVE</code>	Toggles autoDriveActive boolean

LOG ENTRY SLIDE-IN

stiffness: 300,
damping: 28

New log entry animates
from bottom

TELEMETRY NUMBER

duration: 0.6s,
ease curve

animate() interpolates
old→new value

BUTTON PRESS POP

stiffness: 500,
damping: 30

whileTap scale 0.95 on
all buttons

TOAST NOTIFICATION

stiffness: 400,
damping: 20

x: 120→0, scale:
0.85→1

**COMMAND INBOX
SLIDE**

stiffness: 260,
damping: 26

Command arrives from
right edge

GPS PIN MOVEMENT

stiffness: 120,
damping: 18

useSpring tracks
lat/long input changes

**CONNECTED DOT
PULSE**

duration: 2.0s,
loop infinite

scale [1, 1.5, 1], opacity
[1, 0.5, 1]

ERROR SHAKE

stiffness: 800,
damping: 10

High stiffness rapid
oscillation

**LOG SCATTER
(CLEAR)**

stiffness: 100,
damping: 8

Entries fly off screen
before clearing

PANEL COLLAPSE

stiffness: 350,
damping: 30

Device Identity
accordion height:
0↔auto

**CARD GLOW
(TELEMETRY)**

stiffness: 150,
damping: 25

Box-shadow intensity
on value update

TREND ARROW ENTER

stiffness: 180,
damping: 22

y: -4→0, opacity: 0→1
on value change

13.1 Color Palette

TOKEN	HEX VALUE	TAILWIND EQUIVALENT	USAGE
<code>T.bgPrimary</code>	#0d0f12	—	Page background
<code>T.bgCard</code>	#141720	—	All card backgrounds
<code>T.border</code>	#2a2d35	zinc-800	All borders, dividers
<code>T.accent</code>	#f59e0b	amber-500	Primary actions, section headers, active tabs
<code>T.pub</code>	#10b981	emerald-500	Published messages, CONNECTED state
<code>T.sub</code>	#38bdf8	sky-400	Subscribed messages, broker label text
<code>T.error</code>	#f43f5e	rose-500	Errors, alerts, DISCONNECTED button
<code>T.textPri</code>	#e4e4e7	zinc-200	All body text and values
<code>T.textMut</code>	#52525b	zinc-600	Labels, timestamps, secondary info
<code>T.mono</code>	—	—	'JetBrains Mono', 'Fira Code', monospace

13.2 Three-Column Layout

⚡ MQTT DEVICE VIN: T123ZTZT867657777 IMEI: 356741... BROKER STATUS ← 46px Status Bar									
LEFT (300px)		CENTER (flex-1, min-width:0)					RIGHT (350px)		
⊗ Device Identity		↑ PUBLISH MQTT MESSAGES					⊙ LIVE TELEMETRY		
VIN		VTel Loc Events Alert Trip D					Fuel 64% RPM2800		
IMEI							Temp 90° Bat12.6		
TBOX Serial		[Tab content area – forms + buttons]					Odo45230 Lat13.08		
Client ID									
Message ID		↓ COMMAND INBOX							
Corr. ID		[↓ DoorLock] [↓ DoorUnlock]					≡ MESSAGE LOG (n)		
Broker URL		[↓ BlinkerON][↓ BlinkerOFF]					[filters][clear][exp]		
Protocol Ver		[↓ Honk] [↓ FOTA]					[entries...]		
		[↓ OTA] [↓ FetchLog]							
© MQTT CONNECTION		[inbox entries with correlationId]							

14 What Is NOT Implemented (Scope Limitations)

FEATURE	REASON / NOTE	SPEC REFERENCE
X Real MQTT TLS connection	No real broker needed — all simulated in-memory	§6.2, §6.2.6
X Real protobuf binary encoding	Browser security; <code>protobufjs</code> not bundled; mock hex used instead	All proto files
X TLS certificates (<code>ca.crt</code> , <code>client.crt</code> , <code>client.key</code>)	Browser Web Crypto API restrictions on raw TLS socket	§6.2.6
X LWT (Last Will and Testament)	Server-side MQTT broker config — not a dashboard UI concern	§6.2.5
X Exponential retry logic (7 steps, 2s→128s)	Simulated with flat 800ms fake latency	§7.3.2
X Buffered <code>/history</code> topics	<code>/vehicleTelemetry/history</code> etc. not exposed in UI	§8.2
X <code>remoteSMSWakeup</code> flow	Not included in source scripts scope	§10.13
X Diagnostic Alerts topic	Removed from spec in §3.3 update (21-Jan-26) and §4.1	§13.3 (removed)
X Emergency & Safety Alerts	Separate <code>emergencyAlerts</code> topic not in source scripts	§13.4
X FOTA HTTPS download	Binary firmware download is server-initiated, not device UI	§11.2
X <code>userDefinedSpeedCommand</code>	Removed from spec in §4.1 update (06-Feb-26)	§9.5 (removed)
X IMEI Upload / Supplier Feed APIs	REST API flows (not MQTT); backend-only	§7.2.1, §7.2.3

15 How to Use the Dashboard

Quick Start Workflow

STEP	ACTION	EXPECTED RESULT
1	Navigate to <code>/mqtt-virtual-device</code> in Rule Engine	Dashboard loads with dark theme, all 3 columns visible
2	Review VIN/IMEI in Device Identity panel (left col)	Fields pre-filled from .env defaults
3	Click  Generate UUIDs	messageld and correlationId refresh to new UUIDs
4	Click  Save Config	Green toast "✓ Config saved" appears top-right
5	Click  CONNECT	Status → CONNECTING (amber pulse) → CONNECTED (green pulse) in ~800ms
6	Select Vehicle Tel tab → choose signal (e.g., Fuel Level) → click  Publish	Log entry appears in Message Log; Fuel Level card updates
7	Select Location tab → change lat/long → click  Publish Location	GPS pin moves on mini-map; GPS Lat card updates
8	Select Device Join tab → click  Publish deviceJoined	deviceJoined published; 1.2s later a SUB entry for deviceJoinedRsp appears in log
9	In Command Inbox, click  Remote Door Lock	Command slides in with correlationId; click ✓ to send commandResponse
10	Click  Simulate Auto Drive	GPS and telemetry update every 3s; trend arrows activate
11	Click Export JSON in Message Log	Downloads <code>mqtt_log.json</code> with all entries

✔ PUBLISHING PRE-REQUISITE

The dashboard enforces that **you must be CONNECTED before any publish is accepted**. Attempting to publish while DISCONNECTED silently ignores the action (no log entry created). This matches real MQTT behavior.